

In [23]: *# A Pivot table is used to summarize and analyze data by grouping rows a*

```
import pandas as pd
df = pd.read_csv("people-100.csv")
print(df.head())
print(df.columns)
#create Age Column
df['Date of birth'] = pd.to_datetime(df['Date of birth'])
today = pd.Timestamp.today()
df['Age'] = (
    (today - df['Date of birth']).dt.days // 365
)
print(df[['First Name', 'Sex', 'Age']].head())
#basic pivot table
#Average age by gender.
pivot1 = pd.pivot_table(
    df,
    values = 'Age',
    index = 'Sex'
)
print(pivot1)

# pivot table with aggfunc='mean'
pivot2 = pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='min'
)
print(pivot2)
#multiple aggregations
pivot3 = pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc=['mean', 'max', 'min']
)
print(pivot3)
#Add overall average
pivot4 = pd.pivot_table(
    df,
    values='Age',
    index='Sex',
    aggfunc='mean',
    margins=True
)
print(pivot4)
#group by job title and Gender
#Average age by job Title and Gender.
pivot5 = pd.pivot_table(
    df,
    values='Age',
    index='Job Title',
    aggfunc='mean',
    columns='Sex'
)
print(pivot5)
#count people by job and gender
pivot6 = pd.pivot_table(
```

```
df,  
values='Age',  
index='Job Title',  
aggfunc='count',  
columns='Sex',  
fill_value=0  
)  
print(pivot6)
```

	Index	User Id	First Name	Last Name	Sex	\
0	1	88F7B33d2bcf9f5	Shelby	Terrell	Male	
1	2	f90cD3E76f1A9b9	Phillip	Summers	Female	
2	3	DbeAb8CcdfeFC2c	Kristine	Travis	Male	
3	4	A31Bee3c201ef58	Yesenia	Martinez	Male	
4	5	1bA7A3dc874da3c	Lori	Todd	Male	

		Email	Phone	Date of birth	\
0		elijah57@example.net	001-084-906-7849x73518	1945-10-26	
1		bethany14@example.com	214.112.6044x4913	1910-03-24	
2		bthompson@example.com	277.609.7938	1992-07-02	
3		kaitlinkaiser@example.com	584.094.6111	2017-08-03	
4		buchananmanuel@example.net	689-207-3558x7233	1938-12-01	

	Job Title
0	Games developer
1	Phytotherapist
2	Homeopath
3	Market researcher
4	Veterinary surgeon

```
Index(['Index', 'User Id', 'First Name', 'Last Name', 'Sex', 'Email', 'Phone',
```

```
      'Date of birth', 'Job Title'],
      dtype='object')
```

	First Name	Sex	Age
0	Shelby	Male	80
1	Phillip	Female	116
2	Kristine	Male	33
3	Yesenia	Male	8
4	Lori	Male	87

	Age
Sex	
Female	64.981132
Male	54.957447

	Age
Sex	
Female	8
Male	5

	mean	max	min
	Age	Age	Age

Sex			
Female	64.981132	117	8
Male	54.957447	117	5

	Age
Sex	
Female	64.981132
Male	54.957447
All	60.270000

Sex	Female	Male
Job Title		
Accountant, chartered management	95.0	88.0
Accounting technician	17.0	NaN
Advertising account executive	18.0	NaN
Agricultural consultant	NaN	34.0
Air broker	84.0	NaN
...	...	...
Travel agency manager	NaN	117.0
Veterinary surgeon	NaN	87.0
Warden/ranger	NaN	45.0
Warehouse manager	NaN	101.0

Waste management officer	NaN	10.0
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[91 rows x 2 columns]

Sex	Female	Male
Job Title		
Accountant, chartered management	1	1
Accounting technician	1	0
Advertising account executive	1	0
Agricultural consultant	0	1
Air broker	1	0
...	...	...
Travel agency manager	0	1
Veterinary surgeon	0	1
Warden/ranger	0	1
Warehouse manager	0	1
Waste management officer	0	1

[91 rows x 2 columns]